

Long-Yo (Leo) Lee

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GitHub: <https://github.com/ckocklly>

Education

University of Illinois Urbana-Champaign

Aug 2023 – expected May 2027

Bachelor of Science in Mathematics and Computer Science, Minor in Physics

GPA: 3.90/4.00

- **In-Major GPA:** 3.96
- **Dean's List:** Fall 2023, Spring 2024, Fall 2024, Spring 2025
- **Coursework:** System Programming, Artificial Intelligence, Machine Learning, Numerical Analysis, Quantum Computing

Skills

- **Programming Languages:** C/C++/C#, Java/Kotlin, Python, TeX/LaTeX, R
- **Libraries:** pandas, matplotlib, NumPy, pennylane, torch, OpenCV
- **Framework/Tools:** git/GitHub, VS Code, Android Studio, RStudio, Overleaf, Unity
- **Spoken Languages:** Mandarin Chinese, English

Technical and Project Management

Software Engineering Intern

May 2026 – Jul 2026

LuxCreo Inc.

Chicago, IL

- Researched and compiled computer vision algorithms that identifies defects of a membrane used in digital light processing (DLP) 3D printing (github.com/ckocklly/dlp-membrane-defect-detection)
- Built a full-stack application allowing automated bad-bite analysis (github.com/ckocklly/cmsiApp)

Student Researcher | <https://arxiv.org/abs/2510.08501>

Jan 2025 – Oct 2025

Illinois Mathematics Lab, University of Illinois Urbana-Champaign

Urbana, IL

- Rigorously proved concentration inequalities about entanglement measures in quantum information theory
- Implemented simulations using NumPy and PennyLane and determined relations with data-fitting methods
- Represented the research group at the undergraduate poster session of the 2025 MAA MathFest

VR Game Development Intern | <https://github.com/ckocklly/rage-room>

Aug 2024 – Dec 2024

Applied Technologies for Learning in the Arts & Sciences, University of Illinois Urbana-Champaign

Urbana, IL

- Integrated the Oculus XR Plugin for Unity to build immersive, responsive gameplay experiences
- Collaborated in an agile team environment, presenting progress in weekly client meetings
- Delivered a polished demo showcased at the ATLAS final presentation

Audio Lead

Jan 2024 – May 2024

ACM GameBuilders

Urbana, IL

- Composed background music for a video game produced by our club
- Worked closely with the narrative designers to determine the most appropriate musical style for the game theme

Teaching and Tutoring Experience

Course Assistant, CS 173: Discrete Structures

Jan 2025 – present

Siebel School of Computing and Data Science, University of Illinois Urbana-Champaign

Urbana, IL

- Supported a 400-student section by grading proof-based assignments and providing actionable feedback
- Held weekly office hours to help students reason rigorously about logic, counting methods, and algorithms
- Developed the ability to approach students' learning styles by effective communication

Course Assistant, CS 124: Introduction to Computer Science I

Jan 2024 – May 2025

Siebel School of Computing and Data Science, University of Illinois Urbana-Champaign

Urbana, IL

- Supported the 1100-student course by debugging and explaining student code for homework and machine projects.
- Improved instructional materials by designing new practice problems and recording step-by-step walkthroughs
- Mentored and supervised junior course assistants, strengthening leadership and technical review skills

Honor and Awards

Get Experience Scholarship, College of Liberal Arts & Sciences, UIUC	2025
Rank 9, Freshman Mathematics Competition, Dept. of Mathematics, UIUC	2023
Distinction (score: 123/150), American Mathematics Competition 12	2022
Fourth Place Award, International Division, American Regions Mathematics League (ARML)	2022
10/15 (top 5%), American Invitational Mathematics Examination (AIME)	2022